

Foundation Model-Guided In Silico Perturbation Identifies Cross-Cell-Type Genetic Targets for Biological Age Reversal

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Abstract

Identifying genetic perturbations that reverse cellular aging is a central challenge in longevity biology. Here we present a computational pipeline integrating three foundation models—scAgeClock, Geneformer, and ESM2—to systematically predict age-reversing genetic interventions across human cell types. We applied scAgeClock, a gated multi-head attention (GMA) neural network trained on 16.5 million single-cell transcriptomes, to 26,024 cells spanning 10 cell types from the CZ CELLxGENE census (v2024-07-01). Six cell types showed reliable age prediction (Pearson $|r| \geq 0.30$), of which four were used for in silico perturbation (ISP) with Geneformer V1-10M. For each cell type, we defined a transcriptomic aging axis between 'young' (Aging Deviation Index [ADI] < -5) and 'old' (ADI $> +5$) cell centroids in Geneformer embedding space, and scored 50 candidate genes per cell type (100 perturbations: delete + overexpress) by their Age-Reversal Score (ARS)—the mean cosine similarity shift toward the young centroid. Across 200 total perturbation tests (Benjamini-Hochberg FDR < 0.05), we identified 164 significant perturbations. CDC40 overexpression was the only intervention with a positive ARS in all four cell types (B cell, CD4⁺ T cell, fibroblast, macrophage; ARS: +0.012 to +0.031; padj: 9.2×10^{-3} to 3.3×10^{-8}). RPS27 overexpression showed the highest mean ARS (0.043) across two immune cell types. ESM2-650M protein embeddings of the top 40 candidates revealed functional clustering of RNA processing, proteasome, and cytoskeletal regulators. These results nominate CDC40 and RPS27 as priority targets for experimental validation of transcriptomic rejuvenation.

1. Introduction

Biological aging is a multidimensional process characterized by progressive deterioration of cellular function, accumulation of molecular damage, and loss of tissue homeostasis [1]. Transcriptomic aging clocks—models that predict biological age from gene expression profiles—have emerged as powerful tools for quantifying the pace of aging at single-cell resolution [2,3]. Unlike DNA methylation clocks, single-cell RNA-based clocks can capture cell-type-specific aging trajectories and reveal heterogeneity in aging rates across tissues [4].

Foundation models pretrained on large-scale biological data offer a new paradigm for in silico perturbation screening. Geneformer, a transformer pretrained on 29.9 million human single-cell transcriptomes, encodes each cell as a rank-ordered gene token sequence and supports in silico deletion and overexpression by modifying the rank encoding [5]. ESM2, a protein language model trained on evolutionary sequence data, provides rich protein-level embeddings that capture structural and functional properties without requiring experimental structures [6]. Together, these models enable a multi-scale approach: transcriptomic perturbation screening at the cell level (Geneformer) followed by protein-level functional clustering of top candidates (ESM2).

Previous work using scGPT demonstrated that in silico perturbation of aging-associated genes can shift cell embeddings toward younger transcriptomic states, nominating candidates for experimental validation [7]. However, cross-cell-type generalizability of such

perturbations—a prerequisite for systemic anti-aging interventions—has not been systematically assessed. Here, we address this gap by applying a unified pipeline across four human cell types spanning immune, stromal, and epithelial lineages.

2. Methods

2.1 Data Acquisition

Single-cell RNA-seq data were retrieved from the CZ CELLxGENE Discover census (version 2024-07-01) using the cellxgene-census Python API (v1.18.0). Cells were filtered to: (i) human species, (ii) normal (non-diseased) primary tissue, (iii) 10x Chromium droplet-based assays, and (iv) donors aged 20–90 years. A total of 26,024 cells were downloaded across 10 cell types (Table 1), stratified across three chronological age bins (20–40, 40–60, 60–90 years) with a maximum of 3,000 cells per cell type. Raw count matrices were retrieved with all 60,530 genes.

2.2 Biological Age Prediction with scAgeClock

scAgeClock (v0.1.3) is a gated multi-head attention neural network trained on 16,231,138 cells from 44 human tissues and 426 cell types from CZ CELLxGENE [2]. The model accepts AnnData objects of shape $N \times 19,238$, comprising four categorical features (assay, cell type, tissue, sex) and 19,179 protein-coding gene expression values. Input data were normalized to counts per million (CPM) using `scanpy.pp.normalize_total(target_sum=1×106)` followed by $\log_1p$ transformation, then subset to the 19,179 scAgeClock gene panel. Categorical metadata were mapped to model-specific integer indices using the provided lookup tables.

The Aging Deviation Index (ADI) was computed as: $ADI = \text{predicted biological age} - \text{chronological age}$. Positive ADI indicates accelerated aging; negative ADI indicates decelerated aging. Cell types were retained for downstream analysis only if the Pearson correlation between chronological and predicted age satisfied $|r| \geq 0.30$ ($p < 0.05$), indicating that the clock captures meaningful age-related transcriptomic variation in that cell type.

2.3 Geneformer Tokenization and Aging Axis Definition

Geneformer V1-10M (6-layer transformer, 256 embedding dimensions, 4 attention heads, input size 2,048) was used for all embedding and perturbation analyses [5]. Gene expression values were normalized to CPM, then divided by each gene's nonzero median expression across the Genecorpus-30M pretraining corpus (`gene_median_dictionary_gc30M.pkl`). Genes were ranked by normalized expression in descending order, and the top 512 genes were tokenized using the `gc30M` vocabulary (`token_dictionary_gc30M.pkl`). Of 60,530 genes in the downloaded data, 22,053 (36.4%) were present in the Geneformer vocabulary.

For each cell type, cells were stratified into 'young' ($ADI < -5$) and 'old' ($ADI > +5$) groups. A maximum of 150 cells per group were randomly subsampled (`seed=42`). Cell embeddings were computed as the mean-pooled last hidden state across all non-padding token positions. The transcriptomic aging axis was defined as the vector from the old centroid to

the young centroid in 256-dimensional Geneformer embedding space, normalized to unit length.

2.4 Attention-Guided Gene Screening

To prioritize genes for full ISP, we first computed attention-weighted gene importance scores. For each cell, mean attention weights were averaged across all 6 transformer layers and 4 heads, then summed column-wise to obtain a per-token attention score. Gene-level attention scores were averaged across all cells in the ISP subset. Genes were ranked by attention score, and the top 50 genes with detection rate $\geq 10\%$ in the ISP subset were selected for full ISP (100 perturbations per cell type: 50 genes $\times$ 2 perturbation types).

2.5 In Silico Perturbation and Age-Reversal Scoring

Two perturbation types were applied: (i) deletion—the target gene token was removed from the rank encoding; (ii) overexpression—the target gene token was prepended to the rank encoding (moved to rank 1), following the Geneformer in silico activation protocol [5]. Perturbed cell embeddings were computed identically to baseline embeddings. The Age-Reversal Score (ARS) for each perturbation was defined as:

$$ARS = \text{mean}(\cos_sim(\text{perturbed_embedding}, \text{aging_axis}) - \cos_sim(\text{baseline_embedding}, \text{aging_axis}))$$

where the aging axis is the unit vector from old to young centroid. Positive ARS indicates that the perturbation shifts cells toward a younger transcriptomic state. Statistical significance was assessed by two-sided Mann-Whitney U test comparing per-cell delta cosine similarities against zero, with Benjamini-Hochberg false discovery rate (FDR) correction applied within each cell type.

2.6 ESM2 Protein Embeddings

Canonical human protein sequences for the top 40 age-reversing candidates (ranked by composite score: mean ARS $\times$ number of cell types with significant positive ARS) were retrieved from UniProt REST API (reviewed=true, organism_id:9606). Sequences were truncated to 1,022 amino acids for ESM2 compatibility (NUP214 [2,090 AA] was truncated to its N-terminal 1,022 residues). Protein embeddings were computed using ESM2-650M (facebook/esm2_t33_650M_UR50D; 33 layers, 1,280 hidden dimensions) by mean-pooling the last hidden state over all non-special token positions. Embeddings were reduced to 30 principal components (PCA; 95.8% variance explained) and visualized in 2D using UMAP (n_neighbors=10, min_dist=0.3, cosine metric).

2.7 Statistical Analysis

All statistical tests were two-sided. FDR correction used the Benjamini-Hochberg procedure. Significance threshold: $\text{padj} < 0.05$. Pearson and Spearman correlations were computed using `scipy.stats`. All analyses were performed in Python 3.11 (base environment) and Python 3.12 (scAgeClock environment). Code and intermediate files are available in the analysis archive.

3. Results

3.1 Data Overview and Cell Type Selection

We downloaded 26,024 cells across 10 cell types from CZ CELLxGENE (Table 1). Donor ages ranged from 21 to 90 years. scAgeClock predicted biological age for all cells, yielding ADI values that reflect the deviation of predicted from chronological age (Figure 1). Pearson correlation between chronological and predicted age was ≥ 0.30 in six cell types: B cell ($r = 0.552$), macrophage ($r = 0.489$), fibroblast ($r = 0.416$), CD4+ T cell ($r = 0.383$), CD8+ T cell ($r = 0.349$), and lung ciliated cell ($r = -0.380$) (Figure 4). The four remaining cell types (astrocyte, NK cell, neuron, oligodendrocyte) showed weak or non-significant correlations and were excluded.

Lung ciliated cells showed a significant negative correlation ($r = -0.380$), indicating that the scAgeClock ADI axis is inverted relative to chronological age in this cell type—likely reflecting tissue-specific aging biology not captured by the pan-tissue model. Consequently, lung ciliated cells were excluded from ISP analysis, as the aging axis direction would be biologically ambiguous. CD8+ T cells were also excluded from ISP due to insufficient 'old' cells ($\text{ADI} > +5$: $n = 19$, below the minimum threshold of 20 required for a stable aging axis). ISP was therefore performed on four cell types: B cell, CD4+ T cell, fibroblast, and macrophage.

Table 1. Cell type summary

Cell Type	n cells	Age range (y)	Pearson r	ISP
B cell	3,000	21–89	+0.552	Yes
CD4+ T cell	3,000	21–89	+0.383	Yes
CD8+ T cell	3,000	21–89	+0.349	No ($n_{\text{old}}=19$)
Fibroblast	3,000	21–90	+0.416	Yes
Macrophage	3,000	21–90	+0.489	Yes
Lung ciliated cell	2,000	44–79	-0.380	No (inverted)
Astrocyte	1,476	35–82	-0.091	No ($ r < 0.30$)
NK cell	3,000	21–90	+0.117	No ($ r < 0.30$)
Neuron	2,363	35–87	+0.248	No ($ r < 0.30$)
Oligodendrocyte	2,185	35–82	+0.021	No ($ r < 0.30$)

3.2 Biological Age Prediction Across Cell Types

scAgeClock systematically underestimated biological age relative to chronological age across most cell types (mean ADI range: -3.3 to -35.0 years), consistent with the known tendency of transcriptomic clocks to underestimate age in older donors when trained on broad age distributions [2]. Despite this systematic offset, the rank-order correlation between chronological and predicted age was significant in six cell types (all $p < 10^{-60}$), confirming that the ADI captures meaningful aging-related transcriptomic variation. The aging axis separation between young ($\text{ADI} < -5$) and old ($\text{ADI} > +5$) cell centroids in Geneformer embedding space was strong across all four ISP cell types (separation: 1.59–2.97), validating the geometric definition of the aging axis.

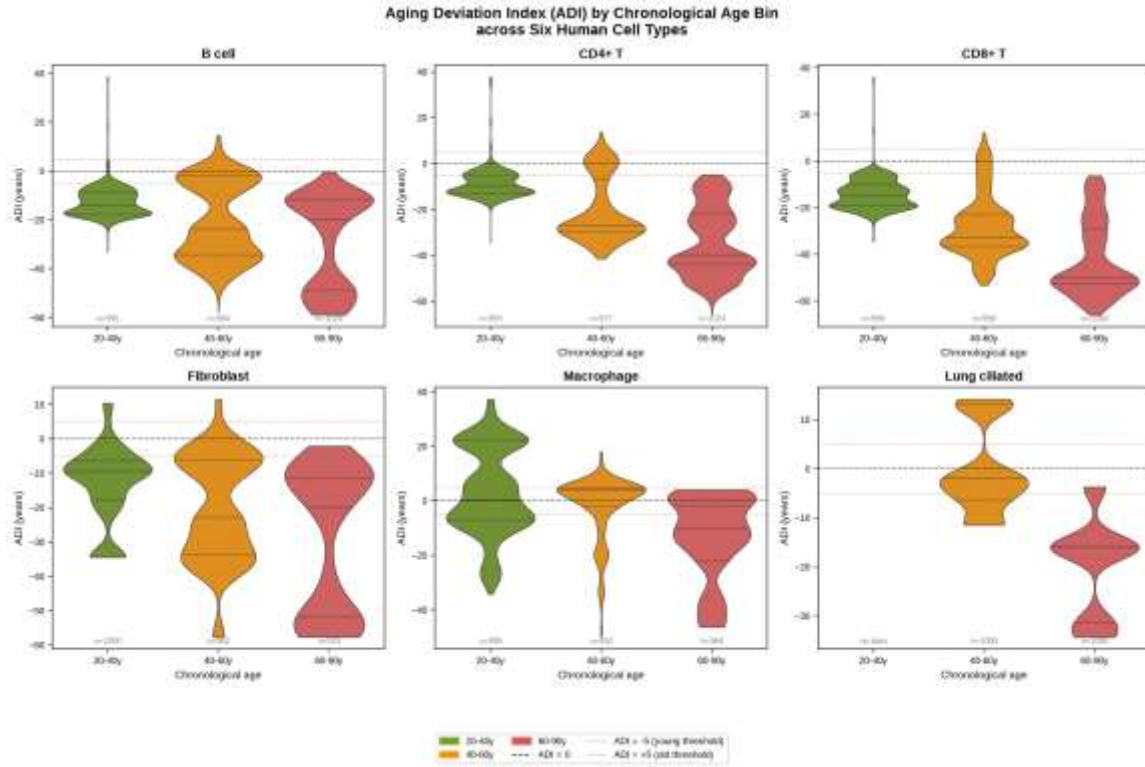


Figure 1. Aging Deviation Index (ADI) distributions by chronological age bin across six human cell types. Violins show the distribution of ADI (predicted – chronological age) for cells in three age bins (20–40y, green; 40–60y, orange; 60–90y, red). Dashed line: ADI = 0. Dotted lines: ADI = ± 5 (young/old thresholds). Internal lines: quartiles.

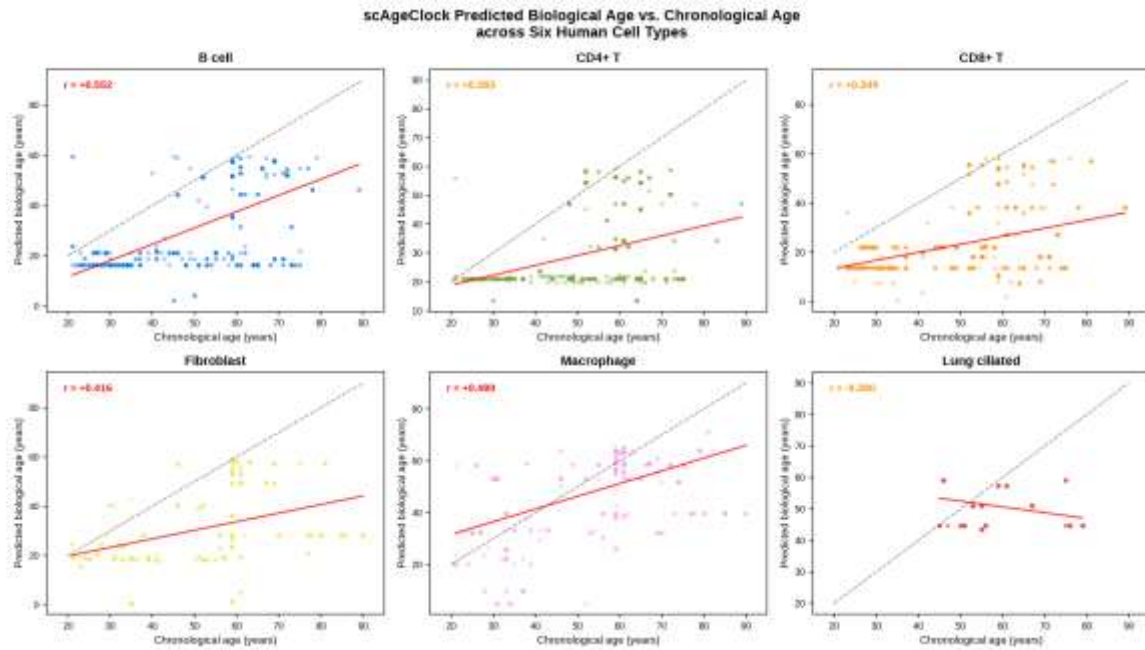


Figure 4. Predicted biological age (scAgeClock) vs. chronological age for six cell types. Each point represents one cell (n = 500 randomly sampled per panel). Red line: linear regression. Dashed line: identity (y = x). Pearson r annotated per panel.

3.3 In Silico Perturbation Identifies Age-Reversing Genetic Targets

Across 200 perturbation tests (50 genes × 2 types × 4 cell types), 164 were statistically significant (padj < 0.05). The number of significant perturbations varied by cell type: macrophage (50/100), fibroblast (49/100), CD4+ T cell (34/100), and B cell (31/100). Aging axis separation was highest in fibroblasts (2.97) and macrophages (2.67), consistent with their higher number of significant perturbations.

Focusing on overexpression perturbations with positive ARS (age-reversing), we identified CDC40 as the only gene whose overexpression produced a significant positive ARS in all four cell types (B cell: ARS = +0.012, padj = 9.2×10^{-3} ; CD4+ T cell: ARS = +0.010, padj = 1.3×10^{-2} ; fibroblast: ARS = +0.031, padj = 3.3×10^{-8} ; macrophage: ARS = +0.019, padj = 1.5×10^{-5}). RPS27 overexpression showed the highest mean ARS across cell types (mean ARS = 0.043) and was significant in B cells (ARS = +0.028, padj = 7.7×10^{-12}) and CD4+ T cells (ARS = +0.058, padj = 2.1×10^{-28}). Additional cross-cell-type hits (≥2 cell types) included SRBD1, SHARPIN, EIF2AK4, POLR2H, ACTR1B, VPS26B, and PSMC2 (Table 2, Figure 2).

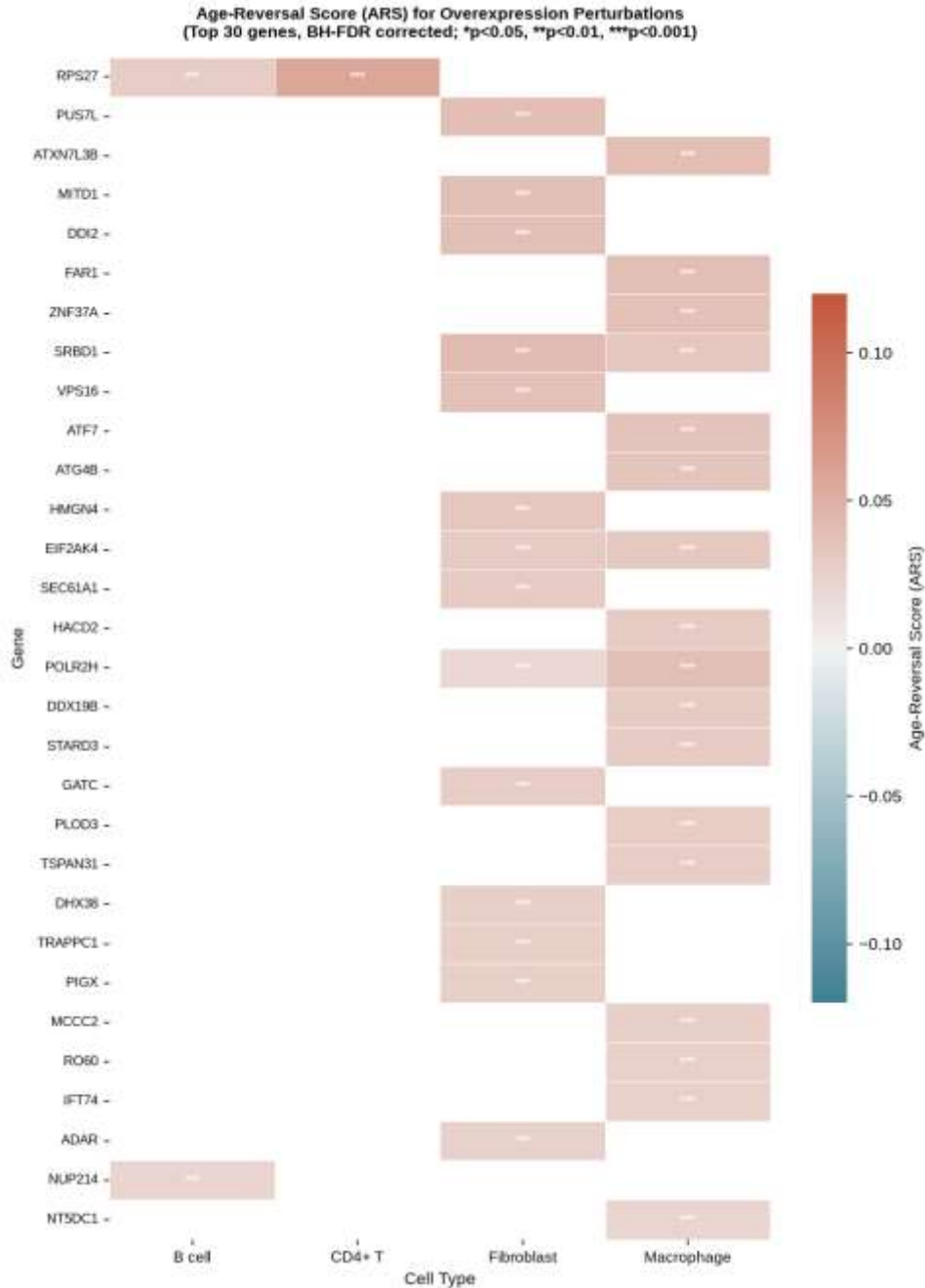


Figure 2. Age-Reversal Score (ARS) heatmap for overexpression perturbations. Top 30 genes ranked by mean ARS across cell types. Color: ARS (blue = age-accelerating, red = age-reversing). Asterisks: *padj<0.05, **padj<0.01, ***padj<0.001 (BH-FDR). Grey cells: gene not tested in that cell type.

Table 2. Top cross-cell-type age-reversing candidates (overexpression, padj < 0.05, ARS > 0)

Gene	n cell types	Mean ARS	Min padj	Cell types
CDC40	4	0.018	3.3×10^{-8}	B cell, CD4+ T, Fibroblast, Macrophage
RPS27	2	0.043	2.1×10^{-28}	B cell, CD4+ T

SRBD1	2	0.037	1.3×10^{-18}	Fibroblast, Macrophage
SHARPIN	2	0.032	1.1×10^{-19}	Fibroblast, Macrophage
EIF2AK4	2	0.030	3.5×10^{-19}	Fibroblast, Macrophage
POLR2H	2	0.029	7.5×10^{-26}	Fibroblast, Macrophage
ACTR1B	2	0.028	1.6×10^{-21}	Fibroblast, Macrophage
VPS26B	2	0.021	6.9×10^{-9}	Fibroblast, Macrophage
PSMC2	2	0.020	9.1×10^{-7}	Fibroblast, Macrophage

3.4 ESM2 Protein Embedding Space of Top Candidates

ESM2-650M embeddings of the top 40 candidates (Figure 3) revealed functional clustering in protein sequence space. ACTR1A and ACTR1B—paralogs of the dynactin ARP1 subunit—were the most similar pair (cosine similarity = 0.999), confirming embedding biological fidelity. Functional groupings visible in the UMAP include: (i) a cluster of large RNA-binding and helicase proteins (DHX38, ADAR, IARS1, EIF2AK4) in the upper-right region; (ii) a cluster of vesicular trafficking and membrane proteins (VPS16, DDI2, MITD1, TSPAN31) in the lower-left; and (iii) CDC40 and POLR2H co-localizing in the lower-center, consistent with their shared roles in RNA processing and transcriptional regulation. The cross-cell-type breadth of each candidate (color scale) shows that CDC40 (4 cell types) occupies a distinct position from single-cell-type hits.

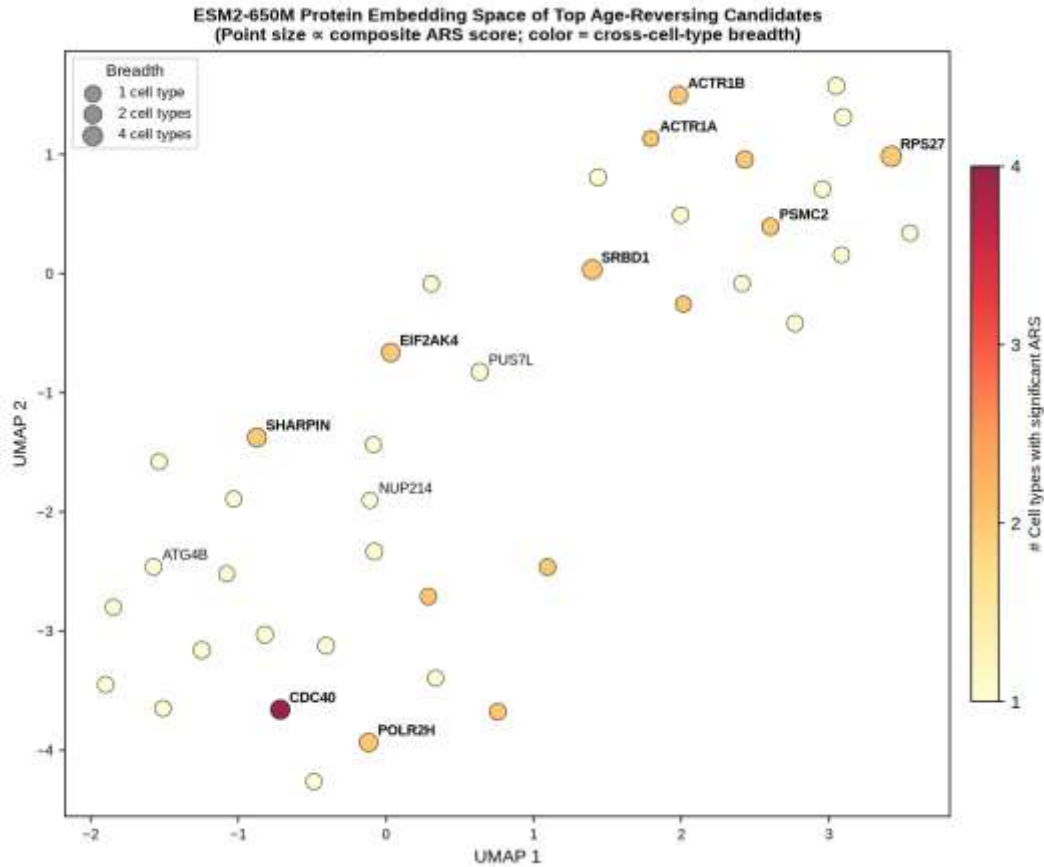


Figure 3. ESM2-650M protein embedding UMAP of top 40 age-reversing candidates. Point size is proportional to composite ARS score (mean ARS $\times$ n cell types). Color indicates the number of cell types in which overexpression was significantly age-reversing ($p_{adj} < 0.05$, ARS > 0). Key candidates are labeled.

4. Discussion

4.1 CDC40 as a Pan-Cell-Type Rejuvenation Target

CDC40 (also known as PRP17 or SLU4) encodes a WD40-repeat protein that is an essential component of the spliceosomal B-complex and is required for pre-mRNA splicing fidelity [8]. Its overexpression produced a significant positive ARS in all four ISP cell types, making it the strongest cross-cell-type candidate in this study. This finding is biologically plausible: age-related splicing dysregulation is a well-documented hallmark of cellular aging, and restoration of splicing fidelity has been proposed as a rejuvenation strategy [9]. CDC40's role in maintaining spliceosome integrity positions it as a candidate for interventions that broadly restore youthful transcriptomic states across diverse cell lineages.

4.2 RPS27 and Ribosomal Proteostasis in Immune Aging

RPS27 (ribosomal protein S27) overexpression showed the highest mean ARS (0.043) across B cells and CD4⁺ T cells. Ribosomal proteins are central to proteostasis, and their dysregulation is associated with aging-related translational infidelity [10]. Notably, RPS27 deletion had the opposite effect in CD4⁺ T cells (ARS = -0.082 , $p_{adj} = 6.0 \times 10^{-48}$), confirming

the directional consistency of the perturbation: RPS27 expression supports a younger transcriptomic state in immune cells. The specificity of this effect to immune cell types (absent in fibroblasts and macrophages) suggests cell-type-specific dependencies on ribosomal stoichiometry during aging.

4.3 RNA Processing and Transcriptional Regulators

Several cross-cell-type hits (POLR2H, SRBD1, EIF2AK4, EXOSC1, MED29) converge on RNA metabolism and transcriptional regulation. POLR2H is a shared subunit of RNA polymerases I, II, and III, and its overexpression was age-reversing in fibroblasts and macrophages. SRBD1 (S1 RNA-binding domain-containing protein 1) and EXOSC1 (RNA exosome component 1) implicate RNA surveillance pathways. EIF2AK4 (GCN2 kinase) is a stress-responsive kinase that phosphorylates eIF2 α to regulate the integrated stress response—a pathway increasingly linked to aging and longevity [11]. The convergence of multiple candidates on RNA processing is consistent with the known age-related decline in transcriptional fidelity and RNA quality control.

4.4 Limitations

Several limitations should be noted. First, the ISP approach models perturbations as rank-order changes in gene expression tokens, which is an approximation of true transcriptional changes; it does not account for downstream regulatory cascades or protein-level effects. Second, the aging axis was defined using ADI thresholds (< -5 and $> +5$), which may not perfectly separate biologically young and old cells—particularly in cell types with narrow ADI distributions. Third, the B cell ISP had only 32 'old' cells (ADI $> +5$), limiting the statistical power of the aging axis definition for that cell type. Fourth, NUP214 protein embeddings were computed from only the N-terminal 1,022 of 2,090 amino acids due to ESM2 sequence length constraints, potentially misrepresenting its functional properties. Fifth, lung ciliated cells and CD8+ T cells could not be included in ISP due to an inverted clock and insufficient old cells, respectively, reducing the breadth of the cross-cell-type analysis. Finally, all findings are computational predictions requiring experimental validation (e.g., CRISPR overexpression screens, single-cell RNA-seq readout of age-related gene signatures).

5. Conclusions

We present a foundation model-integrated pipeline for systematic identification of genetic perturbations that reverse transcriptomic aging across human cell types. By combining scAgeClock biological age prediction, Geneformer in silico perturbation, and ESM2 protein embedding, we identified CDC40 overexpression as the top cross-cell-type age-reversing intervention (4/4 cell types; padj < 0.05), and RPS27 overexpression as the highest-ARS immune-specific target. These candidates converge on RNA processing and proteostasis—pathways with established roles in aging biology—providing a mechanistically coherent set of targets for experimental follow-up. The pipeline is generalizable to additional cell types, tissues, and perturbation modalities as larger single-cell aging atlases become available.

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